

Demonstrating Stepdance: Redesigned CNC Machines Integrating Real-Time Control in Digital Fabrication

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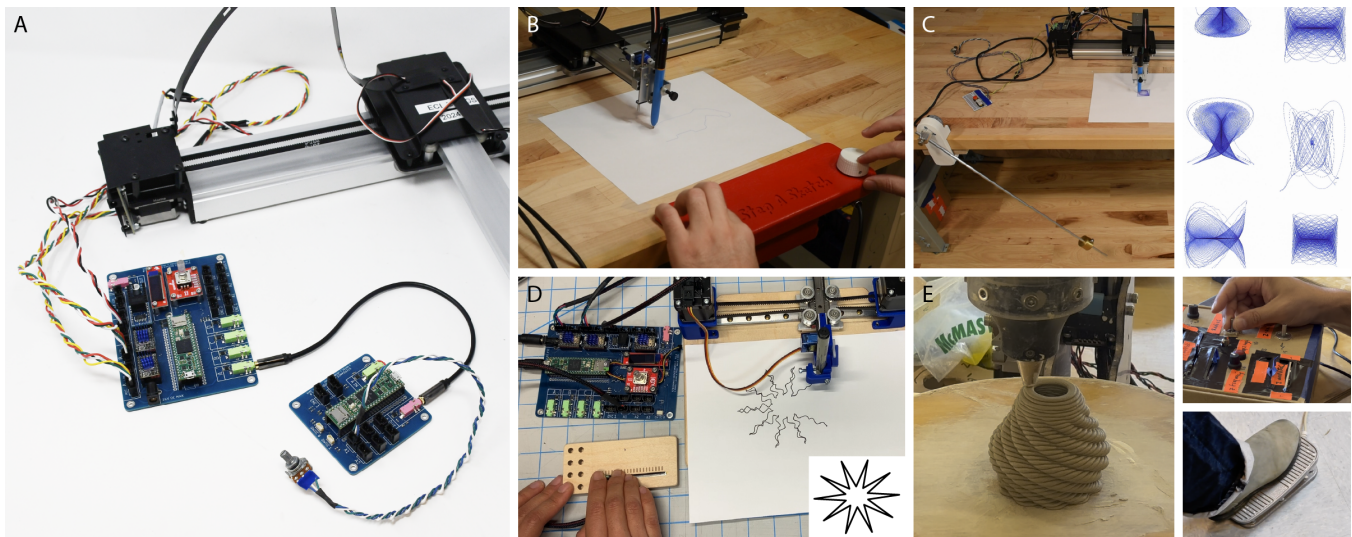


Figure 1: StepDance is a framework for creative CNC motion control. Stepdance hardware modules (A) replace the GCode controller of existing CNC machines and can be chained together to design new physical interfaces for CNC control. We present four redesigned CNC machines that demonstrate how real-time physical input can be integrated within a fabrication workflow. We build these machines by modifying commercial penplotters (B, C, D) and a clay 3D printer (E). We show how real-time control supports playful and simple interactions (B, C), live control over a drawing program (D), and the creation of form and texture via additive fabrication (E).

Abstract

We present a set of redesigned CNC machines for craft-aligned digital fabrication, that combine pre-planned and real-time motion control. These machines re-envision how programmatic machine control can support a diversity of fabrication workflows, in particular demonstrating the potential for machine designers to include

real-time physical input as an interface for users to define the fabricated outcome. We implemented these machines based on a new creative motion control system we developed: Stepdance, which consists of two elements: 1) modular controllers that replace the G-code controller of a CNC and can be chained together to develop new interfaces, and 2) a modular programming library that supports declarative mappings between live user input, pre-programmed operations, and machine motion.

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1 Background

Computer-numerical-control (CNC) machines originated as industrial technology but have since become widely available to individuals and small businesses [Gershenfeld 2005]. Yet, while the applications and users of CNC machines have expanded, the workflows for these technologies have remained largely unchanged from their industrial precursors [Noble 1984]. These standards remain despite practitioners' efforts to establish niche methods [Twigg-Smith et al. 2021] and project-specific workflows [Hirsch et al. 2023]. Nearly all commercial CNC machines require uploading a complete toolpath description, which is then automatically executed by the machine [Twigg-Smith et al. 2021]. The operator is limited to intervening when the machine experiences an error or to performing minor adjustments to machining parameters. While this workflow is well suited to large-scale production, its lack of flexibility hinders more exploratory work, which is essential to the development of innovative materials and products.

2 Objective

We aim to support researchers and designers in developing new machines and interactive workflows to control machines for digital fabrication, in particular exploring the possibility to combine real-time intervention during fabrication, with existing paradigms of pre-planned motion. Doing so has the potential to significantly expand the space of systems, interfaces, and algorithms, developed for computer-aided design and machining workflows in the fields of human-computer interaction, computer graphics, and digital fabrication research. Concretely, we developed StepDance, a motion control system consisting of printed circuit boards (PCBs) (Figure 1A), and a software library, that supports the construction of real-time embedded motion control systems. A key contribution of this system is that it allows using current widely accessible CNC machines as platforms to prototype with, lowering the barrier for researchers to engage with motion control programming.

This Emerging Technology demonstration will give Siggraph attendees the opportunity to interact with four redesigned CNC machines, that integrate physical interfaces for real-time motion control. We built these machines with the StepDance system, by modifying commercial CNC machines (AxiDraw, 3D Potterbot 10 Micro). Beyond a simple demonstration of novel machines, we aim to inspire attendees to imagine new ways to control machines for digital fabrication, while simultaneously providing them an easy way to get started in this space, by releasing our software library and hardware specification under an open-source non-commercial license¹.

3 Redesigned CNC Machines

For an in-depth description of the StepDance system, we refer readers to the full publication [Moyer et al. 2026]. In this section, we describe the four machines that attendees will interact with. These examples showcase a diversity of motion control logic, in particular emphasizing how real-time operator input can be combined with

pre-programmed machine logic to achieve complex and unique outcomes. The three pen-plotter machines present a low-barrier to entry, and enable the attendees to create complete drawings in a very short time, while the clay printer machine showcases a more complex four-axis machine that creates three-dimensional artifacts.

3.1 Pen-plotter as an Etch-a-Sketch

Inspired by popular “Etch-a-Sketch” toy, we modified an AxiDraw pen-plotter [Scientist [n.d.]] to let an operator control the machine's motion in X and Y axes by turning knobs, producing rectilinear drawings (Figure 1B). They can enable or disable drawing by pushing a button to raise and lower the pen. The hardware interface consists of two encoders (for the knobs) and a push button, all wired to the StepDance PCB. The embedded software implementation maps the encoders' rotational motion to translational offsets of the end-effector along X and Y axes, that are then converted to physical stepper motor motion, using the AxiDraw's inverse kinematics equation. This example showcases a simple form of real-time motion control, translating encoder input into stepper motor output. This machine serves as a “hello world” example, helpful to walk attendees through the concept of motion streams².

3.2 Pen-plotter as a Harmonograph

Harmonographs are a class of mechanical drawing instruments where two or more pendulums control the movement of the pen. In a two-pendulum harmonograph, one pendulum moves the pen along the x-axis while the other moves it along the y-axis. Varying the pendulums' frequencies and phases creates different Lissajous curves. We modified an AxiDraw pen-plotter to achieve this effect by attaching two pendulums to the same encoders we used in the previous example, allowing attendees to initiate their swing and observe the resulting drawing (Figure 1C).

3.3 Pen-plotter as a Procedural Artwork

Procedural art is a form of algorithmic art in which the visual outcome is defined through a program and a set of parameters that the artist can tweak to achieve a wide range of outcomes. Procedural art is often combined with pen-plotting (e.g. [noa 2023]), yet physical plotting usually takes place once the entire artwork is generated. With this machine, we demonstrate real-time modification of parameters in a procedural artwork while it is being drawn by the pen-plotter. The operator can use a physical slider to tune the amplitude of a sine wave that is applied onto a pre-programmed SVG path (Figure 1D). Our software implementation additively combines two sources of motion: linear 2D motions corresponding to the SVG path, and sinusoidal motion generated in real-time given the wave amplitude parameter.

3.4 Clay Printer as a Pottery Wheel

Although clay 3D printers use the same material as manual ceramics, they involve a design and interaction process nearly identical to plastic printing. Moyer et al. developed a specialized polar 3D printing mechanism that enables operators to print cylindrical volumes

¹Available at <https://stepdance.org/>

²https://stepdance.org/course/guides/01_motion_streams

in a manner reminiscent of, and compatible with, manual throwing, using a physical foot pedal to control the print bed rotation, and a lever to move the extruder radially [Moyer et al. 2024]. We used StepDance to create a similar interaction on the 3D Potterbot 10 Micro, a widely available commercial machine. We created the hardware interface for the Potterbot wheel by wiring two encoders and a pottery wheel pedal to the machine controller. The embedded software maps pedal input to a polar coordinates angular motion generator that moves the XY bed in a circular motion. Simultaneously, we generate a synchronized Z axis upward motion to achieve spiralized layers, and extruder axis motion to extrude sufficient clay, proportionally to the distance traveled by the nozzle. This creates a cylindrical form with varying radius. Lastly, we added the possibility to add surface texture effects on the form during the print (Figure 1E), by combining the base cylindrical motion with a 1D sine wave generator with a tunable amplitude, frequency and phase (through a set of physical potentiometers). This last example invites visitors to experience a completely different workflow for 3D printing: one in which the form is not fully defined digitally as in traditional CAD and CNC workflows, but continuously changed and updated as the print progresses.

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